SATYA SREEJA CHALASANI

AI & CLOUD SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

AI and Cloud Engineering professional with **7+ years of experience** across software development, full-stack engineering, cloud operations, data engineering, and AI application development. Skilled in building full-stack applications using **React.js, Angular, TypeScript, Python, FastAPI, Flask, and PostgreSQL**. Experienced in developing **Generative AI applications, LLM-powered solutions, RAG pipelines, and agentic AI workflows** using LangChain, LangGraph, OpenAI API, Gemini API, FAISS, and ChromaDB. Hands-on experience with **AWS, Microsoft Azure, and GCP**, including cloud infrastructure, data processing, Docker, Terraform, and CI/CD automation. **AWS Certified Solutions Architect – Associate**, with experience in application deployment, infrastructure provisioning, monitoring, testing, and troubleshooting.

TECHNICAL SKILLS

Frontend Development	React.js, Angular 15, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Angular Reactive Forms, RxJS, Routing, Route Guards, Angular Services, Directives, Pipes
Backend Development	Python, FastAPI, Flask, REST API Development
AI / Generative AI	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, LangGraph, OpenAI API, Gemini API, Hugging Face, Prompt Engineering, Embeddings
Vector Databases & Search	FAISS, ChromaDB, Semantic Search, Vector Indexing
AWS Cloud	EC2, S3, IAM, VPC, Subnets, Security Groups, CloudWatch, AWS CLI, Application Load Balancer (ALB), Auto Scaling Groups (ASG), Lambda, API Gateway, RDS, DynamoDB, ECS Fargate
Microsoft Azure	Virtual Machines, Azure Storage, Azure Monitor, Networking
Google Cloud Platform (GCP)	Google Cloud Dataproc, PySpark
DevOps & Infrastructure	Docker, Terraform, Infrastructure as Code (IaC), Git, GitHub, GitHub Actions, CI/CD, Linux, Windows
Databases	PostgreSQL, SQLite, SQL
Testing & API Tools	Pytest, Postman, Swagger / OpenAPI, Unit Testing, Integration Testing
Monitoring & Troubleshooting	Splunk, Application Logging, Incident Management

PROFESSIONAL EXPERIENCE

AI & Cloud Software Engineer

Jan 2025 - Present

Client: CBRE, Texas, USA

- Developed Python-based AI application services using **FastAPI** and REST APIs, integrating **OpenAI API** and **Gemini API** for natural-language processing, contextual question answering, content generation, and information extraction.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines covering document ingestion, text extraction, chunking, embedding generation, vector storage, and similarity-based retrieval using **FAISS** and **ChromaDB**.

- Developed multi-step **agentic AI workflows** using **LangChain** and **LangGraph**, integrating LLMs, external tools, and application services; worked with **Hugging Face** models and optimized prompts and retrieval strategies to improve response relevance.
- Developed interactive and reusable frontend components using **React.js, JavaScript, TypeScript, HTML5, CSS3, and Tailwind CSS**, integrating them with backend AI services through REST APIs.
- Built backend integrations using **Python, SQL, and PostgreSQL** to support application logic, data retrieval, processing, and storage.
- Containerized Python applications using **Docker** and worked in **Linux** environments to manage application processes, execute shell commands, review logs, and troubleshoot deployment issues.
- Deployed and supported applications on **AWS EC2**, configuring **IAM permissions, security groups, VPCs, subnets, and networking** to enable secure application access and connectivity.
- Managed **Amazon S3** buckets and access permissions for cloud storage and application data management; used **AWS CLI** to manage resources, verify configurations, and support deployment troubleshooting.
- Monitored application and infrastructure activity using **Amazon CloudWatch**, reviewing logs and investigating deployment-related issues.
- Configured **Application Load Balancers (ALB)** and **Auto Scaling Groups (ASG)** to support load distribution, health checks, and scalable application deployments.
- Worked with **AWS Lambda, API Gateway, Amazon RDS, DynamoDB, and ECS Fargate** to support serverless integrations, relational and NoSQL data storage, and containerized application workloads.
- Provisioned and managed AWS infrastructure using **Terraform (Infrastructure as Code)** and maintained version control and CI/CD workflows using **Git, GitHub, and GitHub Actions**.
- Implemented application logging, exception handling, and troubleshooting procedures; performed API validation, documentation, and integration testing using **Postman, Swagger/OpenAPI, and Pytest**.

Technology Stack: *Generative AI, LLMs, RAG, Agentic AI, OpenAI API, Gemini API, LangChain, LangGraph, Hugging Face, Prompt Engineering, Embeddings, FAISS, ChromaDB, React.js, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Python, FastAPI, REST APIs, PostgreSQL, SQL, Linux, AWS EC2, S3, IAM, VPC, Subnets, Security Groups, CloudWatch, ALB, ASG, Lambda, API Gateway, RDS, DynamoDB, ECS Fargate, AWS CLI, Terraform, IaC, Docker, Git, GitHub, GitHub Actions, CI/CD, Postman, Swagger/OpenAPI, Pytest.*

SOFTWARE ENGINEER - CLOUD & APPLICATION DEVELOPMENT

Client: Salesforce, Texas, USA September 2022 - July 2024

- Developed interactive and reusable frontend components using **Angular 15, JavaScript, and TypeScript**, with **HTML5, CSS3, and Tailwind CSS** for responsive application interfaces.
- Implemented **Angular Reactive Forms, RxJS, Angular services, routing, route guards, custom directives, and pipes** to support form validation, asynchronous operations, application navigation, and maintainable frontend functionality.
- Integrated frontend components with backend **REST APIs**, handling API requests and responses, user interactions, application state, and dynamic data rendering.
- Developed backend services using **Python, FastAPI, and Flask** to implement business logic and expose application functionality through REST APIs.
- Implemented database operations using **PostgreSQL and SQL**, including data retrieval, insertion, updates, and validation.
- Used **Postman and Swagger/OpenAPI** to inspect API endpoints, validate request and response structures, and troubleshoot API integration issues.

- Performed **unit and integration testing using Pytest** to validate backend functionality, API behavior, and interactions between application components.
- Supported application deployments in **AWS**, including **EC2-based hosting**, security group configuration, and access permissions.
- Used **Docker** to containerize applications and worked with **Git and GitHub** for source-code management and collaborative development.
- Used **GitHub Actions** to support **CI/CD workflows** involving automated builds, testing, and deployment; investigated defects across frontend components, APIs, backend logic, and database connectivity.

Technology Stack: *Angular 15, JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, Angular Reactive Forms, RxJS, Angular Services, Routing, Route Guards, Directives, Pipes, Python, FastAPI, Flask, REST APIs, PostgreSQL, SQL, AWS EC2, Docker, Git, GitHub, GitHub Actions, CI/CD, Postman, Swagger/OpenAPI, Pytest, Unit Testing, Integration Testing.*

Software Developer

Apr 2021 - Aug 2022

Client: First Databank, California, USA

- Developed interactive and reusable frontend components using **React.js, JavaScript, HTML5, and CSS3**, integrating them with backend REST APIs to retrieve, submit, and display application data.
- Developed **Python scripts and backend application components** to support business logic, data retrieval, application integrations, and data-processing workflows.
- Used **Google Cloud Platform (GCP), Google Cloud Dataproc, and PySpark** to process, clean, transform, and standardize structured datasets for downstream application workflows.
- Implemented **data validation and quality checks** to identify missing values, duplicate records, inconsistent formats, and data discrepancies.
- Wrote **SQL queries** and performed database operations using **PostgreSQL and SQLite**, including data retrieval, filtering, joins, insertion, updates, and record validation.
- Integrated backend functionality through **REST APIs** and used **Postman and Swagger/OpenAPI** to test endpoints, validate request parameters, inspect response payloads, and troubleshoot integration issues.
- Performed **functional and integration testing** and used **Pytest** to validate Python functionality, database operations, data transformations, and API responses.
- Investigated application errors and data discrepancies by reviewing application logs, reproducing issues, and validating corrective changes.
- Used **Git and GitHub** for source-code management and collaborated with cross-functional teams to resolve technical issues and support application enhancements.

Technology Stack: *React.js, JavaScript, HTML5, CSS3, Python, SQL, PostgreSQL, SQLite, REST APIs, Data Processing, Data Validation, Google Cloud Platform (GCP), Google Cloud Dataproc, PySpark, Backend Development, Git, GitHub, Postman, Swagger/OpenAPI, Pytest, Unit Testing, Functional Testing, Integration Testing.*

Software Developer

Dec 2018 - Mar 2021

Client: ZF Group, Michigan, USA

- Developed and maintained **Python-based application components and automation scripts** to support business logic, data processing, and repetitive validation activities.

- Built basic frontend components using **JavaScript, HTML5, and CSS3**, integrating them with backend **REST APIs** to support application functionality and data exchange.
- Wrote **SQL queries** and performed relational database operations, including data retrieval, filtering, joins, record updates, and database-level validation.
- Performed application debugging and root-cause analysis to identify backend errors, data discrepancies, and functionality issues.
- Executed **unit, functional, and integration testing** and used **Postman** to validate API functionality, troubleshoot service communication, and verify application behavior.
- Supported cloud infrastructure operations using **Microsoft Azure**, including provisioning and managing **Azure Virtual Machines**, performing storage operations through **Azure Storage**, and monitoring resource health using **Azure Monitor**.
- Assisted with **VM provisioning and decommissioning, snapshot management, disk operations, and resource configuration** to support infrastructure requirements.
- Worked with **Windows Server** environments on Azure Virtual Machines and supported **Azure networking configurations**, including Network Interface Cards (NICs), virtual network connectivity, and connectivity troubleshooting.
- Monitored VM health and resource alerts using **Azure Monitor** and used **Splunk** to review application logs, investigate errors, troubleshoot system issues, and support root-cause analysis.
- Supported **incident management activities** by investigating infrastructure and application incidents, analyzing alerts and logs, coordinating issue resolution, and documenting troubleshooting actions.
- Used **Git and GitHub** for source-code version control and collaborated with developers and QA engineers to investigate defects, validate fixes, support software releases, and document technical changes and test results.

Technology Stack: *JavaScript, HTML5, CSS3, Python, SQL, Relational Databases, REST APIs, Backend Development, Microsoft Azure, Azure Virtual Machines, Azure Storage, Azure Monitor, Azure Networking, Windows Server, NICs, Splunk, Incident Management, Git, GitHub, Postman, Unit Testing, Functional Testing, Integration Testing.*

CERTIFICATIONS

AWS Certified Solutions Architect - Associate
Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
Microsoft Certified: Azure Fundamentals (AZ-900)
Anthropic Certified: Introduction to MCP, AI Fluency, Claude 101

EDUCATION

M.S. Data Science | University of North Texas
Bachelor's in Data Science | Loyola Academy Degree & PG College